

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

FIRST SEMESTER 2014-2015

SURVEYING (COMPREHENSIVE EXAM) REGULAR - MODEL SOLUTIONS

Course No: CE F213

Date: 01-12-2014 (Monday)

Duration: 3 hours

Max. Marks: 80 (Weight: 40%)

I: Choose the best answers/Fill up the blanks

[10 x 1 = 10]

- 1) A 30 m chain is generally divided into **150** links and each link measures **20** cms
- 2) Chain line is represented by _____
- 3) FAR stands for **Floor Area Ratio**
- 4) The ideal transition curve is **clothoid**
- 5) An offset making an angle with the chain line other than 90° is termed as **oblique offset**
- 6) A first sight taken on a levelling staff held at a point of known elevation is termed as **back sight**
- 7) Reduced level of a point is the elevation with reference to **datum**
- 8) Lemniscate transitional curves are generally provided along **highways**
- 9) Total deflection angle of a transition curve is equal to **1/3** of the spiral angle.
- 10) Contours cannot intersect but may unite to form a single contour (**True/False**).

II: Short answers:

[10 x 2 = 20]

- 1) What is an index sketch?

Index sketch is prepared on the basis of reconnaissance survey and is pasted on the first page of field book for reference.

- 2) What is a 12 cm compass?

The diameter of the compass box is 12 cms.

- 3) What are the methods of plane tabling?

Radiation, intersection, traversing, resection.

- 4) Define super elevation. Why is it required?

Rising of outer edge of the road/railway with respect to inner edge. It is required to prevent the vehicles from running off the track.

- 5) What is meant by "length of a vertical curve"?

It is the horizontal distance between two points of tangencies.

- 6) Differentiate between carpet area and built-up area.

Carpet Area: Actual usable area; It does not even include the thickness of walls.

Built-up area: It includes the area of the walls also.

7) What is the difference between Bowditch rule and Transit rule?

Bowditch rule: Bowditch rule is employed when lines and angular measurements of the traverse are of equal accuracy. Hence, corrections are proportional to distances only.

Transit Rule: Angular measurements are more accurate than the linear measurements. Hence, corrections are proportional to latitudes rather than distances.

8) Define Most Probable Value.

Most Probable Value is the value which is most likely be the true value.

9) What is meant by photo-theodolite?

Photo-theodolite is the instrument in which theodolite is fitted with camera to take photographs. Mostly adopted in photogrammetry.

10) Differentiate between error, mistake and discrepancy.

Error: Difference between the measured value and true value. It can be systematic or accidental. Error can be due to the limitation of the instrument in measuring certain value.

Mistake: Mistake is more of due to negligence, carelessness.

Discrepancy: It is the difference in the value of two like measurements.

Small discrepancy does not mean that error is small.

LONG ANSWERS:

1(a) Vol. between LDP & 70 m; 80% of 450 = 360 hectares

$$V = \frac{3}{2} [7+4] = 16.5 \times 10^4 \text{ cu.m} \quad [1]$$

Let TWL be 90m;

$$V_{70-90} = \frac{5}{2} [(23+7) + 2(17+15+13)] = 300 \times 10^4 \text{ cu.m} < 360 \text{ hectares} \quad [1]$$

Let TWL be 95m;

$$V_{70-95} = \frac{5}{2} [(25+7) + 2(23+7+15+13)] = 420 \times 10^4 \text{ cu.m} > 360 \text{ hectares}$$

$$\text{Additional vol. required} = \frac{[300+16.5]}{\text{Acquired vol}} - \frac{260}{\text{Actual volume reqd.}} = 43.5 \text{ hectares} \quad [1]$$

$$V_{\text{reqd}} = \frac{x}{2} [23 + 23 + \frac{2x}{5}]$$

$$43.5 = \frac{x}{2} [46 + \frac{2x}{5}] \Rightarrow x = 1.86 \text{ m} \quad [2]$$

$$\text{TWL for 80\% storage capacity} = 90 + 1.86 = 91.86 \text{ m.}$$

1(b)	Line	Latitude	D.M.D	Lat x DMD
	AB	+225.5	120.5	27172.75
	BC	-245.0	451.0	-110495.00
	CD	-150.5	551.0	-82850.25
	DA	+170.0	220.0	37400.00

$$\text{Algebraic Sum} = -128,772.5 = \text{Twice the area}$$

$$\text{Required area of traverse} = \frac{1}{2} \times 128772.5 = 64,386.25 \text{ m}^2$$

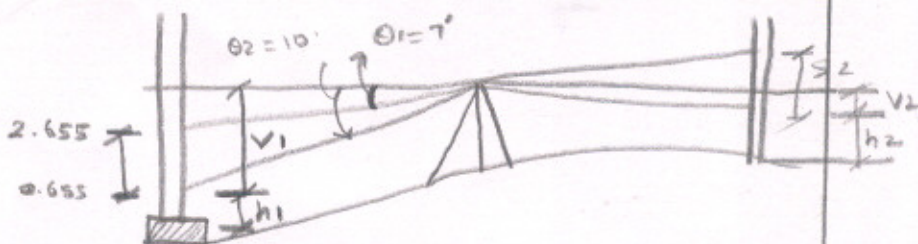
Marking scheme:- DMD: 3 marks; Lat x DMD: 0.5 marks; Final area: 1.5 marks

2(a) First observation:-

A to BM:-

$$S_1 = 2 \text{ m}; h = 0.655 \text{ m};$$

$$\theta_1 = 7'; \theta_2 = 10'$$



$$D_1 = \frac{S_1}{\tan \theta_2 - \tan \theta_1} = \frac{2}{\tan 10' - \tan 7'} = 37.354 \text{ m}$$

$$V_1 = D_1 \tan 10' = 37.35 \times \tan 10' = 6.586 \text{ m} \quad [1]$$

$$\begin{aligned} \text{RL of inst. axis} &= \text{RL of BM} + h_1 + V_1 = 510.500 + 0.655 + 6.586 \text{ m} \\ &= 517.741 \text{ m} \quad [1] \end{aligned}$$

Second observation:-

A to B:-

$$h_2 = 1.250 \text{ m}; S_2 = 1.95 \text{ m};$$

$$\theta_1 = 5'; \theta_2 = 4'$$

$$D_2 = \frac{S_2}{\tan \theta_1 + \tan \theta_2} = \frac{1.95}{\tan 5' + \tan 4'} = 12.388 \text{ m}$$

$$V_2 = D_2 \times \tan 5' = 1.084 \text{ m} \quad [1]$$

$$\text{RL of B} = \text{RL of inst. axis} - V_2 - h_2 = 517.741 - 1.084 - 1.250 = 515.407 \text{ m.} \quad [2]$$

2(b) $\angle A = 44' \quad \angle B = 26'30' \quad \angle C = 159'45' \quad \angle D = 42'15' \quad \angle E = 267'30' \quad [1]$

Line	F.B	B.B	
AB	194'15'	14'15'	[1 mark]
BC	40'45'	220'45'	[1 mark]
CD	20'30'	200'30'	[1 mark]
DE	242'45'	62'45'	
EA	330'15'	150'15'	[1 mark]

NOTE: If answer is found using other method, Then 2.5 marks will be awarded.

3) Radius = $402.633 \text{ m} \left[\frac{v^2}{9R} = \frac{1}{8} \right]$ [2 marks]

$L_{\text{trans}} = \frac{V^3}{Rc} = 90.82 \text{ m}$ [2 marks]

Shift = $\frac{L_{\text{tran}}^2}{24R} = 0.854 \text{ m}$ [0.25 marks]

$L_{\text{Tang}} = (R+s) \tan \Delta/2 + \frac{L}{2} = 278.36 \text{ m}$; [0.25 marks]

$L_{\text{circ}} = \frac{\pi R (\Delta - 2\phi)}{180} = 330.82 \text{ m}$ [0.5 marks]

Chainage at $T_1 = 1150 - 278.36 = 871.64 \text{ m}$; chainage at F.T.P = $871.64 + 90.82 = 962.46 \text{ m}$ [0.5]

Chainage at S.P = $962.46 + 330.82 = 1293.28 \text{ m}$ [0.5]

Chainage at $T_2 = 1293.28 \text{ m} + 90.82 = 1384.1 \text{ m}$ [0.5 marks]

Setting out of a Circular curve. Setting out of a transition curve:

$O_n = \frac{C_n [C_n + C_{n-1}]}{2R}$ [1.5 marks]

or

$S_n = \frac{1718.9 C}{R}$

$\alpha = \frac{573 L^2}{RL} \text{ minutes}$ [1.5 marks]



4(a)

$\tan \alpha_1 = \frac{h_1}{d} \rightarrow (1)$

$\frac{h_1}{d} = \frac{h_1 + 1.5}{d + 15} \Rightarrow d = 10h_1$

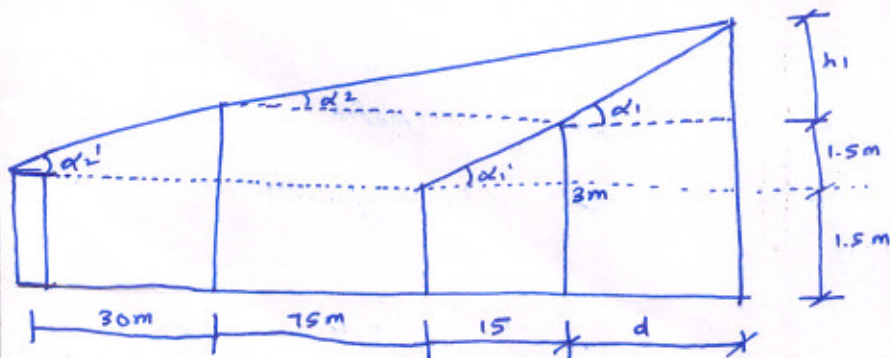
$\tan \alpha_2 = \frac{h_1}{d + 90} \rightarrow (3)$

$\tan \alpha'_1 = \frac{h_1 + 1.5}{d + 15} \rightarrow (2)$

$\tan \alpha'_2 = \frac{h_1 + 1.5}{d + 120} \rightarrow (4)$

$\frac{h_1}{d + 90} = \frac{h_1 + 1.5}{d + 120}$

$h_1 = 9 \text{ m}$



$\therefore \text{Height of tower} = 9 + 1.5 + 1.5 = 12 \text{ m}$ [5 marks]

NOTE:- one more answer is possible when you keep two ranging rods on opposite sides of a tower.

4(b) True difference in level = Mean of apparent differences in level

$$= \frac{(2.810 - 2.165) + (2.355 - 0.910)}{2}$$
$$= 1.545 \text{ m} \quad [2 \text{ marks}]$$

Error in collimation = -0.004 m in 100 m ;

$$\text{Error in } 1530 \text{ m} = \frac{-0.004}{100} \times 1530 = -0.0612 \text{ m}$$

$\therefore$ Correction for collimation = $+0.0612 \text{ m}$

$$\left. \begin{array}{l} \text{Combined correction for curvature} \\ \text{refraction} \end{array} \right\} = -0.0673 D^2$$
$$= -0.0673 (1.53)^2 = -0.1575 \text{ m.}$$

When the instrument is kept at "A", staff readings at "A" are assumed to be correct;

Correct staff reading at B should be $(\overset{2.165}{\cancel{2.165}} + 1.545) = 3.710$

$$(3.810 - 3.710) = 0.1 \text{ m} \quad [\text{Total error}]$$

$$\therefore 3.810 - C_c + C_r + C_{\text{coll}} = 3.710$$

$$3.810 - 0.0785(D)^2 + C_r + 0.0612 = 3.710$$

$$\Rightarrow 3.810 - 0.184 + C_r + 0.0612 = 3.710$$

$$\Rightarrow \boxed{C_r = 0.02256} \quad [\text{Best way to determine } \overset{\text{refraction}}{\text{curvature}} \text{ correction}] \quad [3 \text{ marks}]$$

NOTE:- Alternatively: $C_r = \frac{1}{7} \times C_c = \frac{1}{7} \times 0.184 = 0.02628$

5 Remote Sensing:- Process of obtaining information about a particular object without being in actual physical contact with the object. [2]
space-based; Air borne... [2]

Total station:- Electronic theodolite + distance measuring device. [1]

No, Total station can't measure horizontal distance. [1]

GPS:- Global positioning system [1] Can't be used indoors; Requires 4 satellites to establish a point. Atomic clocks $\rightarrow$ very precise to establish time.... [1]

GIS:- Geographical Information system; Extensively used in all civil engineering applications... [0.5]